GreenLight Research

An agentic research assistant for healthcare papers, that also tells you if your idea is new.

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> **One line:** A researcher asks a question or describes an idea. A team of AI agents finds the relevant healthcare papers, answers questions from them with citations, keeps fetching more literature when needed, and scores how trustworthy each paper is. On top of that, **GreenLight** judges whether the idea is already done (**green**: open, **amber**: partly done, **red**: done) and points to new topics worth researching.

1. The problem

- **Finding and reading the right papers is slow.** A researcher in healthcare and life sciences searches several databases, opens dozens of PDFs and still is not sure the important work was found.
- **Getting answers out of papers is manual.** "Which methods were tried for X? On which patients? With what results?" means reading paper after paper and taking notes.
- **Not every paper deserves trust.** Weak baselines, tiny samples, over-claimed results and retracted work exist. Search ranks by relevance, not reliability.
- **"Has this already been done?" has no quick answer.** If a topic is saturated, months of work add little. If it is barely touched, there is a real opening. Today that judgement takes weeks.

Who feels this most: PhD students and early researchers, lab heads and grant writers, and R&D teams scanning a field.

2. Core: the research assistant

Capability	What the researcher gets
Paper discovery	Ask a research question; agents search healthcare sources in parallel and return the most relevant papers, preprints and trials, with links.
Ask the papers	To-and-fro chat that answers questions *from the papers themselves* (retrieval over abstracts and full texts). Every claim carries a citation to the paper and passage it came from.
Fetch more literature	When the collected papers cannot answer a question, the agent says so and goes back to search for more, then answers again.
Credibility score	Every paper gets a 0–100 trust score from how often and how well it is cited, how reputed its authors are, where it was published, and verified red flags. Retracted papers score zero.
Evidence check	Drafted answers are checked against the source text; unsupported or contradicting claims are flagged.
Research graph	An interactive map of papers and researchers: who cites whom, which papers are similar, which clusters exist and where the empty spaces are.
Memory	The chat remembers the whole session, so the researcher can dig deeper, compare papers and change direction.

3. Add-on: GreenLight, is this idea new?

Capability	What the researcher gets
Idea intake	A short chat turns a rough idea into a clear brief: question, method, population and intended contribution.
Novelty matrix	The idea's components against the closest papers. An overlap only counts if the paper's own words prove it.
For-and-against debate	A Proponent agent and a Skeptic agent argue the case with evidence; a judge agent rules green, amber or red. Fixed rules can only make the verdict stricter.
New topics to research	Instead of just "already done", it finds under-explored angles (a new population, method or outcome) and confirms each with a targeted search, so the researcher leaves with concrete directions.

4. Healthcare sources we follow

- **PubMed:** peer-reviewed biomedical literature.
- **Europe PMC:** abstracts, open full texts and publication counts per year (how crowded a topic is).
- **medRxiv and bioRxiv:** newest preprints, clearly marked as not peer-reviewed.
- **ClinicalTrials.gov:** is someone already running a trial on this?
- **Cochrane systematic reviews:** the strongest "already done" signal.
- **WHO, FDA and EMA:** guidelines and regulatory context.
- **OpenAlex and Semantic Scholar:** citations, author reputation and reference lists for the credibility score and the graph.

5. How it works

- **A team of agents, each with one job:** planner, lead researcher, parallel scouts, critic, reader, credibility auditor, answer writer, evidence checker, and for GreenLight a novelty mapper, proponent, skeptic and judge.
- **Deep research loop:** the lead splits a question into angles and sends scouts out in parallel; a critic rejects thin coverage and forces another round.
- **Context compression:** long searches are folded into short digests and a working memory, and the lead sees only the scouts' compact reports. Long research sessions stay fast and focused.
- **Shared notebook:** agents write findings to a common blackboard every agent can read.
- **Live view:** the researcher watches the agents work in real time: who is searching what, which papers were found, what was rejected.

6. Why researchers can trust it

- **Every answer is cited.** Each claim links to the paper and passage it came from, and unsupported claims are marked.
- **Says "not enough evidence" instead of guessing.** If the papers do not support an answer, it fetches more or says so.
- **Credibility is computed, not guessed.** The trust score comes from citation data, author reputation, venue and verified red flags, not from an AI's opinion.
- **Claims about papers need the paper's own words.** Overlaps and red flags carry a quote checked against the text.
- **The verdict is guardrailed.** If a credible paper already covers every core part of the idea, the light is red whatever the debate says. If the search was thin, it is never green.

7. Existing tools and how we differ

Tool type	What they do well	Gap
Literature assistants (e.g. Elicit, Consensus)	Find and summarise papers that answer a question	No trust score per paper; do not judge whether *your idea* is new
Citation tools (e.g. Scite)	Show whether citations support or dispute a claim	Per-claim view; no research chat or novelty verdict
Graph tools (e.g. Connected Papers, ResearchRabbit)	Visualise related literature	Exploration only; no answers, trust score or decision

Our difference: one place to find papers, ask them questions with citations, know which ones to trust, and decide whether an idea is worth pursuing, with new directions when it is not. Built for healthcare first. *(Comparison from general public knowledge, not an in-depth review.)*

> **The story in one breath:** Researchers lose weeks finding papers, reading them for answers and wondering which ones to trust, and months on ideas that are already done. GreenLight Research finds the papers, answers from them with citations, scores their credibility, and tells the researcher whether the idea is new and where the real openings are.